

LAB REPORT: String Handling in C++

Title: String Length Calculation and String Concatenation in C++

1. String Length Calculation

Objective

To write a C++ program that declares and initializes a string and uses a built-in function to calculate its length.

Theory

In C++, strings are objects of the `string` class.

The `length()` or `size()` function is used to find the number of characters in a string.

It also includes spaces and special characters.

Program Code

```
#include <iostream>
#include <string>
using namespace std;

int main() {

    // Declare and initialize string
    string message = "Welcome to C++ Programming Lab";

    // Calculate length using built-in function
    int length = message.length();

    // Display result
    cout << "Message: " << message << endl;
    cout << "Length of the string: " << length << endl;

    return 0;
}
```

Explanation (Step by Step)

- `string message` → a variable is created to store text

- "Welcome to C++ Programming Lab" → assigned value
 - `message.length()` → counts total characters in the string
 - `cout` → displays output on screen
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Expected Output

Message: Welcome to C++ Programming Lab
Length of the string: 31



Conclusion

The program successfully calculates the number of characters in a string using the built-in `length()` function.



2. String Concatenation



Objective

To write a C++ program that combines two strings (first name and last name) into a full name.



Theory

String concatenation means joining two or more strings together. In C++, the `+` operator is used to concatenate strings.



Program Code

```
#include <iostream>
#include <string>
using namespace std;

int main() {

    // Declare and initialize strings
    string firstName = "Aziz";
    string lastName = "Ahmad";

    // Concatenate strings
    string fullName = firstName + " " + lastName;

    // Display result
```

```
cout << "First Name: " << firstName << endl;
cout << "Last Name: " << lastName << endl;
cout << "Full Name: " << fullName << endl;

return 0;
}
```

Explanation (Step by Step)

- `firstName` stores first name
 - `lastName` stores last name
 - `+` operator joins both strings
 - `" "` adds a space between names
 - `fullName` stores final combined result
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Expected Output

```
First Name: Aziz
Last Name: Ahmad
Full Name: Aziz Ahmad
```

Conclusion

This program demonstrates how to combine two strings using the concatenation operator in C++.

Final Summary

Task	Concept Used	Function/Operator
String Length	Counting characters	<code>length()</code>
String Concatenation	Joining strings	<code>+</code> operator